

Fit Over Depth

Transfer-Appropriate Compression for Long-Context LLM QA

“Longer context means the model loses track.”

This is the common belief about long-context LLM QA — is it actually true?

Background: “Long-context QA” means feeding an LLM an entire document (up to tens of thousands of words) and asking it to answer questions about any part of it — no separate retrieval step, no summarization. The intuition is that somewhere in that huge input, the model “loses the thread.” This is often called the *Lost-in-the-Middle* problem.

- A recent study links this failure to proactive interference (similar information overwriting itself in memory) — and shows it tracks model size, not context-window length.
- This looks structurally like the capacity limits of human working memory.
- That parallel is the whole reason this talk borrows from cognitive psychology.

The belief doesn't hold — but a real problem does

+1.4%p

net accuracy change after padding a 10K-word document to 50K words (143 fixed-answer questions)

Length-stress test: 4 genres, 607 questions, 1,000–50,000 words

The real problem: a flat “utilization gap”

News genre — the model has almost no prior knowledge (closed-book accuracy ≈ 0):

Even with the **full document given directly**, accuracy never exceeds 0.4.

The answer is right there in the text — the model still misses most of it.
So the failure is not about length. It's about *using* what's already in front of the model.

What to look at: the +2 stat shows length isn't the enemy; the news-genre box shows the real, length-independent gap this paper targets.

BACKGROUND

LLM pipelines already borrow from human memory — just not on purpose

Three prior systems each reinvent a piece of human memory research, in isolation.

Prior work	What it does	Human-memory parallel
C-DIC (dialogue compression)	Keeps multiple “thread” summaries, updates the closest match	Elaborative rehearsal — restating info by linking it to context
ReadAgent	Reads a gist first, re-fetches the original page if needed	Reconstructive memory — gist first, details on demand
Adaptive-k	Picks how many passages to retrieve, per question	(used later, for retrieval sizing)

No one has tested these memory strategies side by side, in one pipeline, to see which ones actually transfer to LLMs — and why.

Three memory strategies, in plain terms

Elaborative Rehearsal

Restate information by linking it to what you already know.

e.g. *Like rewriting your notes in your own words after class.*

Testing Effect

Practice retrieving information from memory, rather than re-reading it.

e.g. *Like using flashcards instead of re-reading the textbook.*

Maintenance Rehearsal

Repeat information as-is, with no reorganizing.

e.g. *Like repeating a phone number in your head until you dial it.*

Two theories about which strategy should win

Levels-of-Processing

Craik & Lockhart, 1972

“Deeper” processing (elaborative, meaning-based) always beats “shallower” processing (maintenance, surface repetition) for long-term memory.

→ **Predicts a single ranking, always the same order.**

Transfer-Appropriate Processing

Morris, Bransford & Franks, 1977

No single depth wins. What matters is the match between how information was encoded and how it will be retrieved.

→ **Predicts the winner depends on the task/document.**

What to look at: these two theories make opposite predictions — the results later in this talk will adjudicate between them.

Three research questions guide this talk

RQ1 Do human memory-strengthening strategies actually help an LLM use long documents better?

RQ2a

Does rewriting content vs. preserving it decide whether a strategy works at all?

RQ2b

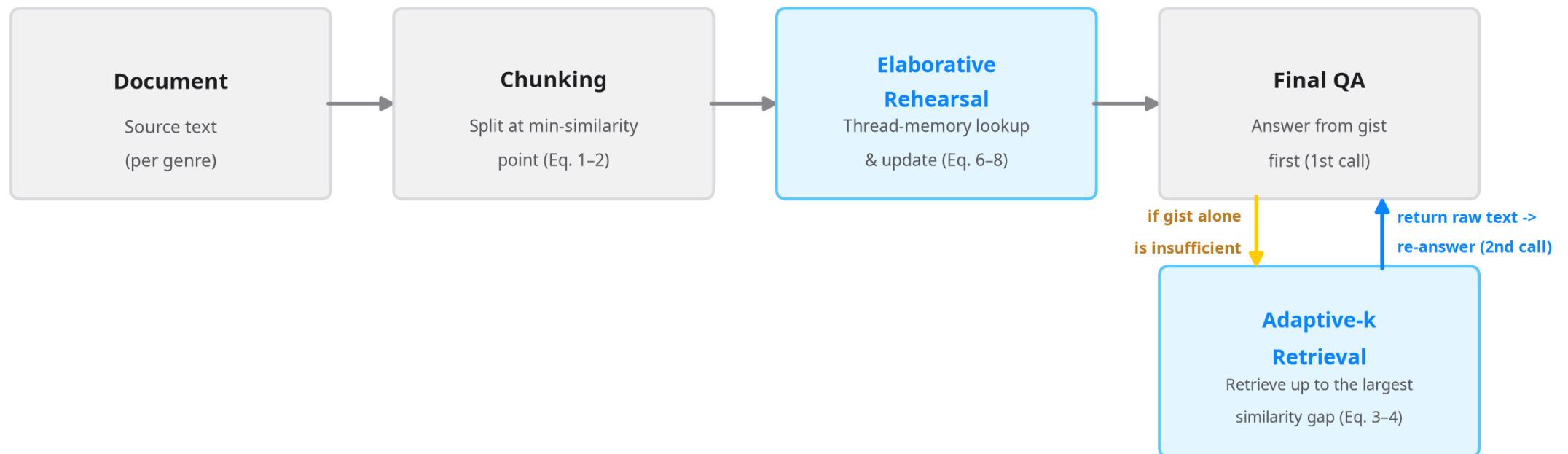
Among strategies that survive, does how well they fit the document's structure decide which one wins?

Preview: the first strategy tested — elaborative rehearsal — is rejected by RQ2a. The paper's real answer to RQ1 comes from two other strategies, explained by RQ2b.

The proposed pipeline: three stages



Modules trained/built in this study



What to look at: the blue-outlined boxes are the modules this project trained; gray boxes are fixed pre/post-processing steps.

Chunking: splitting at natural breakpoints

- The document is split wherever two adjacent units of text (paragraphs or sentences) are least similar in meaning.
- Sentence and dialogue-quote boundaries are always respected first.
- Author-marked scene breaks override everything else.

$$s_i = \cos(e_i, e_{i+1}) \quad i^* = \arg \min_i s_i$$

e_i = embedding of unit i . s_i = similarity to the next unit.

The boundary

i^* is wherever that similarity is lowest, inside a min/max word-count window.

What to look at: the drop in the similarity curve is the cut point — this is the same idea as classic TextTiling, extended with embeddings (Eq. 1–2).

Elaborative rehearsal: what it does, how it's trained

- Reads chunks in order; each step folds the current chunk into a running summary — it rewrites, not just appends.
- Stage 1: supervised one-shot compression (XSum data) on t5-small.
- Stage 2: retrained on the actual rolling, multi-thread process, using a 70B teacher model's demonstrations (336 training examples).
- The compressor is deliberately small (t5-small) — humans need rehearsal because working memory is limited; giving the compressor unlimited capacity would beg the question.

Two-stage training

- 1 One-shot compression (XSum, supervised)
- 2 Rolling / threaded (70B teacher demos)

Training objective: predict the teacher's summary

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \log P_{\theta}(S_i \mid C_i, S_{i-1})$$

θ	the compressor's trainable parameters
C_i	the current chunk being read
S_{i-1}	the running summary so far
S_i	the teacher's updated summary — what the model is trained to reproduce

What to look at: this is just negative log-likelihood — 'make the model's summary match the teacher's summary as closely as possible.'

Thread memory: which summary gets updated?

The model keeps several running summaries (“threads”) — one per topic — and updates the one closest to the current chunk.

$$\delta_t = \max_j \text{sim}(C_t, Z_j) \quad j^* = \arg \max_j \text{sim}(C_t, Z_j)$$

δ_t = highest similarity found; j^* = which thread it belongs to

If $\delta_t < \tau$ (no close thread)

→ open a new thread for this topic

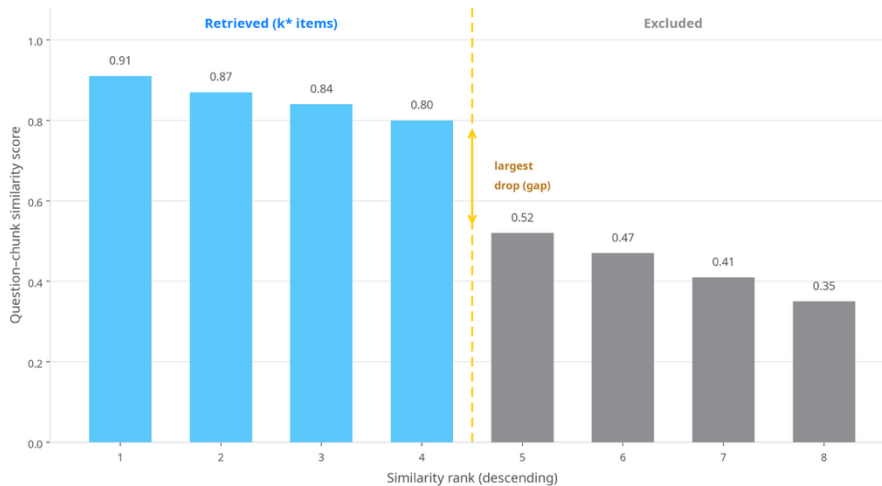
If $\delta_t \geq \tau$ (a close thread exists)

→ replace that thread with the updated summary

(Eq. 5 — a capacity cap on thread count triggers merge/evict/grow, so memory doesn't grow forever.)

Final QA: answer from gist first, retrieve if needed

- First call: answer using every chunk's gist (the running summaries).
- If insufficient, the model names which chunk(s) to re-fetch verbatim (“Lookup”) for a second call.
- Which chunks to fetch is decided by Adaptive-k: rank chunks by similarity to the question, then cut at the sharpest drop in that ranking — no hand-tuned count per genre.



What to look at: bars 1–4 (blue) are kept; the dashed line marks the biggest score drop — everything past it (gray) is dropped.

Validated on 4 genres, 579 questions

Baselines: Closed-Book, Full-Context. Compared: elaborative-rehearsal Lookup / Lookup+Retrieval, and pure retrieval-augmented generation (RAG).

Genre	Source	# Questions	Scoring
Wiki	SQuAD	100	EM / F1
News	NewsQA	200	EM / F1
Fiction	NarrativeQA	79	EM / F1
Case law	CaseHOLD (LexGLUE)	200	5-way accuracy

RAG here means no learned compression at all — just Adaptive-k selection over the raw original text.

At full scale, plain RAG beats elaborative rehearsal

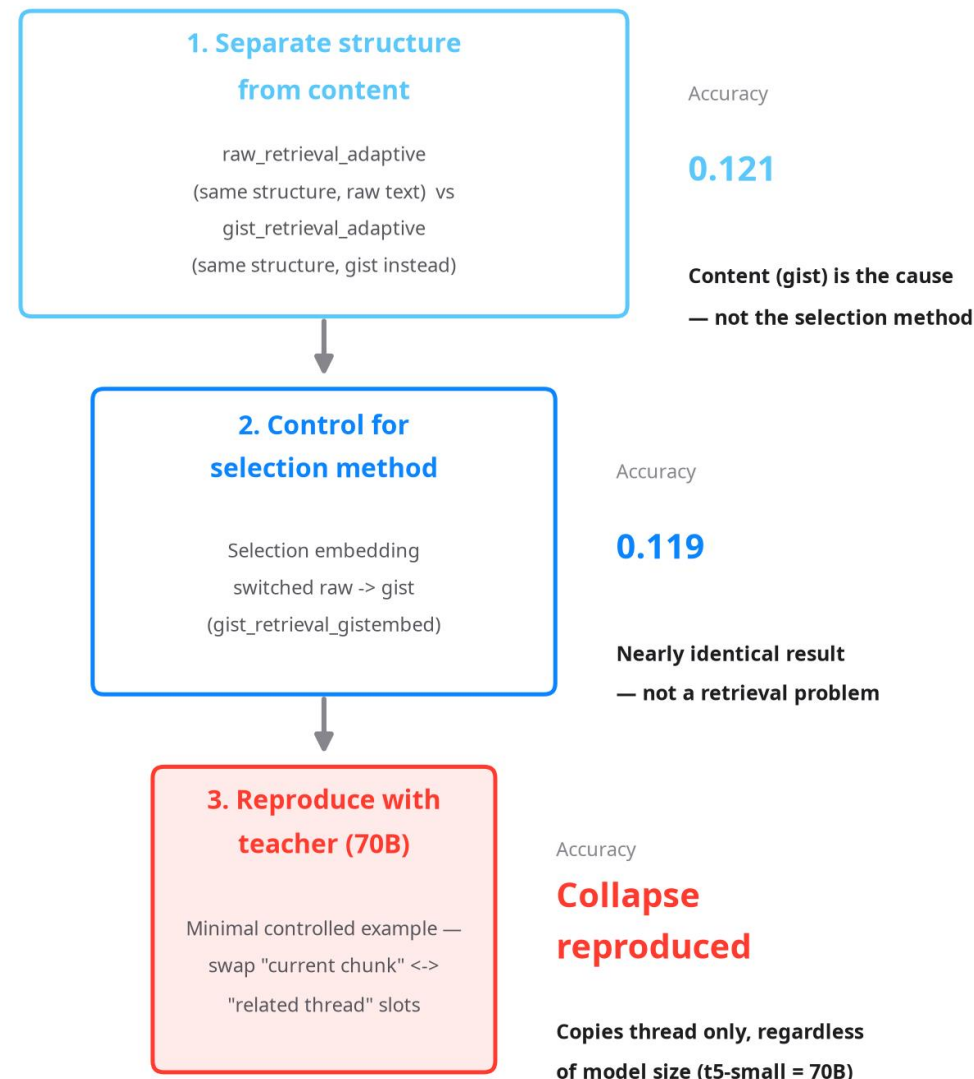
The pattern seen at pilot scale (15 questions/genre) didn't flip when the sample grew to 579 — it got sharper.

Condition	Wiki	News	Fiction	Case law	Overall
Full-Context (entire document)	0.810	0.395	0.494	0.360	0.515
Lookup+Retrieval (best rehearsal)	0.620	0.220	0.291	0.245	0.339
RAG	0.850	0.420	0.316	0.300	0.439

What to look at: the blue row (RAG) wins on every genre, with no training and no compression at all — the question becomes: why does rewriting lose?

Narrowing the cause in three steps

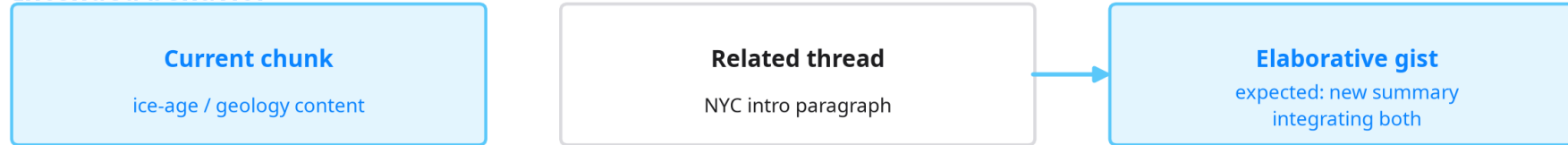
- Step 1 — same selection structure, content swapped for gist: accuracy drops to 0.121. Selection isn't the problem.
- Step 2 — selection embeddings switched to gist too: 0.119, almost identical. Retrieval isn't the problem either.
- Step 3 — reproduced with the 70B teacher on a minimal controlled example: same collapse, independent of model size.



-> Thread-memory collapse — a failure of the retrieve-and-replace design itself, independent of model size

The compressor copies the wrong slot

Intended behavior



Actual observed behavior (collapse)



What actually happens: the model reproduces the retrieved “related thread” almost verbatim and ignores the “current chunk.” Swapping which slot holds which content reproduces the same failure — this is a structural bug in retrieve-and-replace, not a data or scale issue.

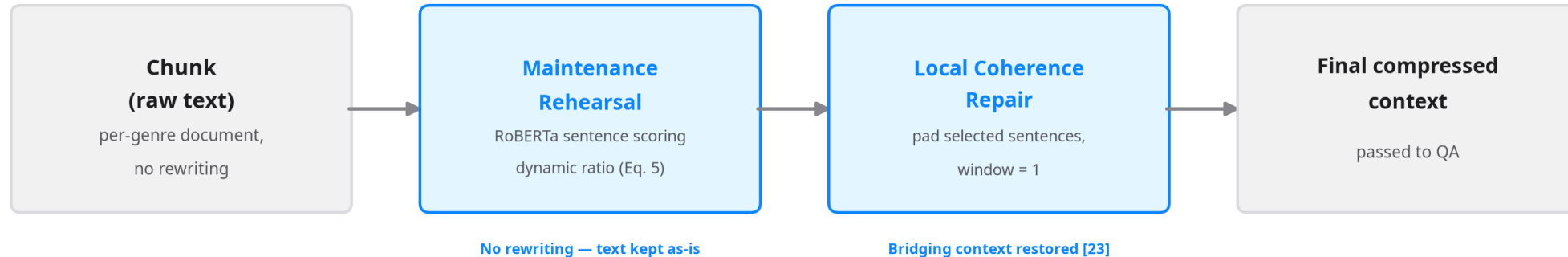
The Proposed Model

Transfer-Appropriate Extraction + Local Coherence Repair

Only strategies that don't rewrite can survive the collapse just diagnosed. This section builds the model from that constraint upward.

The proposed model, end to end

Proposed Model = Transfer-Appropriate Extraction (no rewriting) + Local Coherence Repair



254 / 254

matches RAG's exact answer count (out of 579 questions)

-14% tokens

avg. 3,081 vs RAG's 3,588 — equal or fewer in every genre

What to look at: no rewriting anywhere in this chain — maintenance rehearsal selects, local coherence repair pads. That's the whole mechanism.

Maintenance rehearsal + a length-adaptive compression ratio

- Maintenance rehearsal: pick important sentences within a chunk with a RoBERTa scorer, keep them verbatim — no cross-chunk merging, so the retrieve-and-replace collapse structurally cannot occur.
- Fixed ratio alone already helps: $R = 0.3 \rightarrow 0.5$ raises accuracy $0.325 \rightarrow 0.368$ (fewer evidence sentences get lost, 43% $\rightarrow$ 26%).
- Best result: ratio set dynamically per document length (Eq. 8, target budget $K = 32,000$ tokens).

$$R = \min\left(1, \frac{K}{|D|}\right), \quad K = 32,000$$

R = compression ratio kept; $|D|$ = document length in tokens; K = fixed token budget

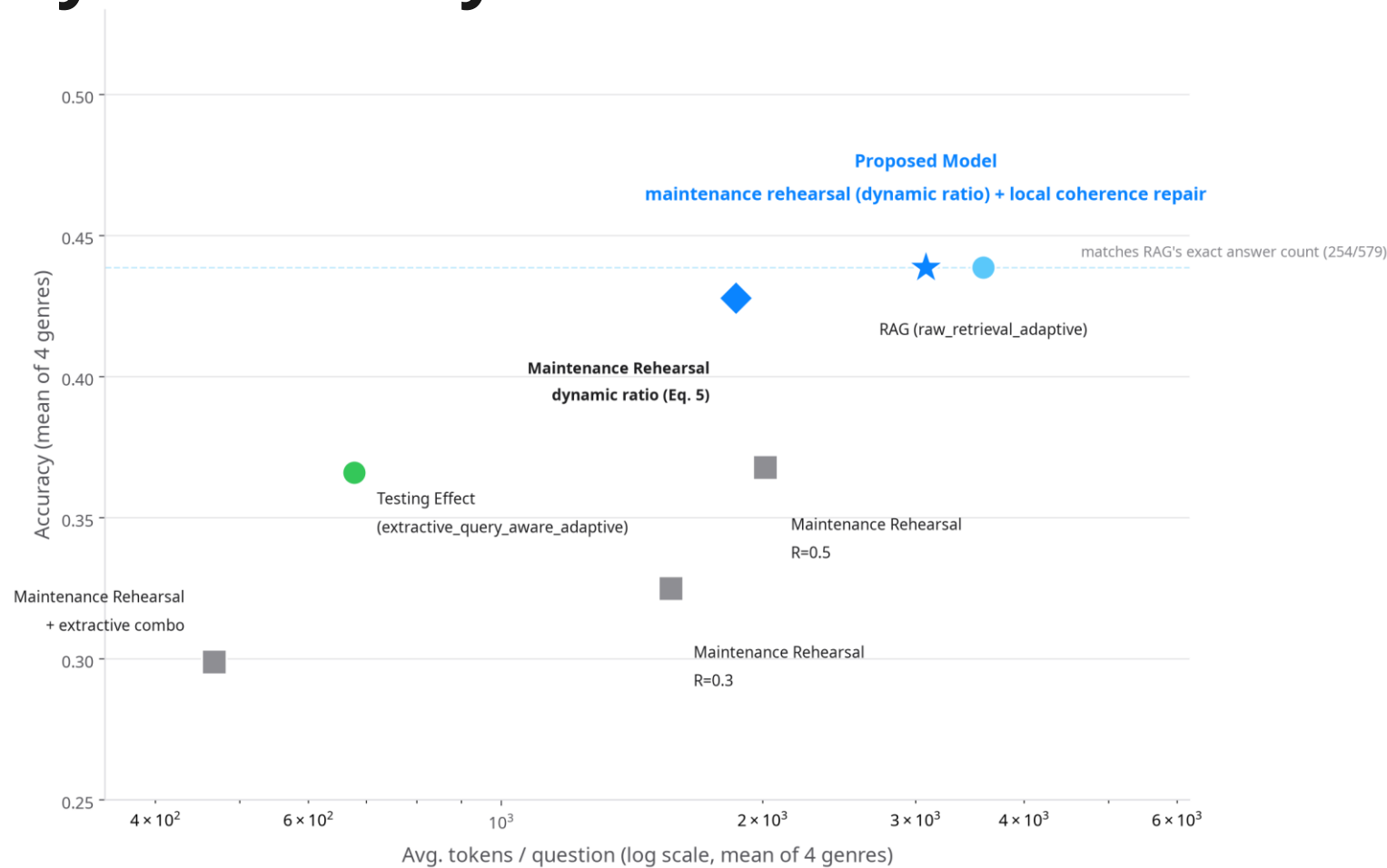
0.428

overall accuracy — within 0.011 of RAG (0.439),
at half RAG's tokens (1,862 vs 3,588)

0.880

Wiki accuracy — beats RAG's 0.850, the only
condition to exceed RAG outright

The efficiency frontier: dynamic ratio dominates fixed ratio



What to look at: diamonds (dynamic ratio) sit above gray squares (fixed ratio) at similar token cost; the star (final model, next section) reaches RAG's circle with fewer tokens.

What's left: extraction creates “seams”

Fiction still trails RAG (0.278 vs 0.316) — genres that depend on narrative continuity are hit hardest.

Why this happens: extraction stitches together sentences that weren't adjacent in the source — that stitch is a seam. Kintsch's (1988) Construction-Integration model treats comprehension as building local links between propositions; missing links force the reader into *bridging inference* (Clark & Haviland, 1977). In narrative specifically, Zwaan, Langston & Graesser's (1995) event-indexing model tracks five continuity threads readers monitor automatically:

Time

Space

Character

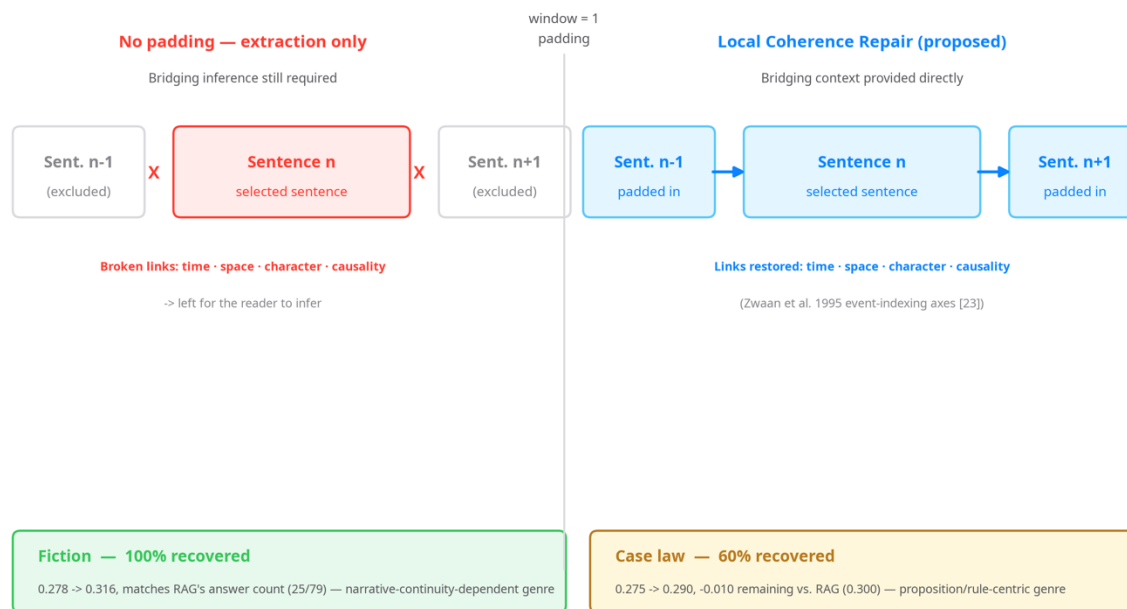
Causality

Intention

Structured, less narrative genres (case law) depend on these axes less — so they should lose less to extraction. That asymmetry is testable, not just a post-hoc story.

Fix: pad one sentence on each side

- Keep the extracted sentence, plus one sentence immediately before and after it (window = 1) — the bridging context comes back for free.
- Fiction (79 questions): gap fully closed — 0.278 → 0.316, matching RAG's exact answer count (25/79), using fewer tokens (858 vs 977).
- Case law (200 questions): 60% of the gap closed — 0.275 → 0.290.



Predicted in advance, not fit after the fact: narrative genres depend more on the five continuity axes (S21), so they should recover more — and they do (100% vs 60%).

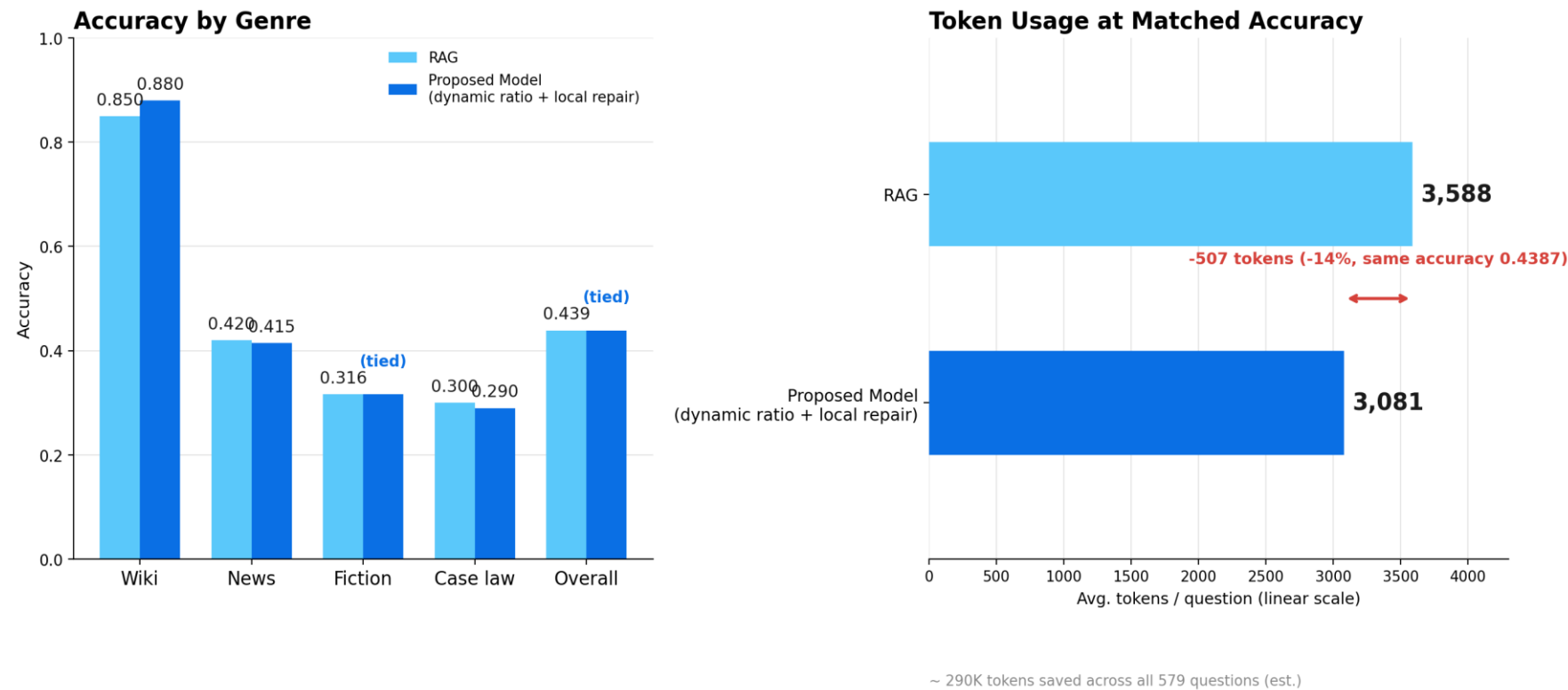
Resolving the two theories from earlier

Levels-of-Processing predicted deep (elaborative) always wins — the opposite happened, because elaboration's rewriting step was structurally broken (RQ2a), not because depth is wrong in general.

Genre	Document shape	Winner	Gap vs. RAG
Wiki	short	Maintenance rehearsal (dynamic ratio)	+3.0%p (exceeds RAG)
Fiction	long, narrative-dependent	Maintenance rehearsal + repair = RAG	Testing effect alone: −20.2%p
News / Case law	medium, low narrative	RAG ≈ Maintenance rehearsal	−0.5%p to −1.0%p

Among strategies that don't rewrite, the winner still changes by genre — exactly what Transfer-Appropriate Processing predicts (RQ2b).

Final model vs. RAG: same answers, fewer tokens



254 / 254
correct answers out of 579 — exact match with RAG

3,081 vs 3,588
average tokens — 14% fewer than RAG, in every genre

0.4387
overall accuracy — identical to RAG to 4 decimal places

Answering the three research questions

RQ1

Do memory strategies help LLMs use long documents?

Yes — but not the strategy assumed at the outset.

RQ2a

Does rewriting vs. preserving decide if a strategy works?

Yes — rewriting (elaborative rehearsal) collapses structurally, independent of model size.

RQ2b

Among survivors, does fit-to-structure decide the winner?

Yes — the winner changes by genre, matching Transfer-Appropriate Processing.

Limitations and what's next

Limitations

- Table 2's numbers are a single-run point estimate — no significance testing yet.
- Case law's re-retrieval reversal is now diagnosed: Adaptive-k finds the evidence chunk only 7% of the time in a 400-chunk shared space (200 unrelated cases). Bypassing retrieval: 0.290 -> 0.560.
- The full 3×2×2 factorial design (both rehearsal strategies × combination) wasn't run — testing-effect + maintenance-rehearsal in sequence is untested.

Future work

- Validate the correction-retry mechanism at full scale.
- Apply local coherence repair to the testing-effect (extractive) condition to close its remaining gap.
- Use teacher-model gist quality to establish an upper bound on compressor quality.

Three things resolved this week

Resolved

- Case law's targeting failure, confirmed: Adaptive-k finds the evidence chunk only 7% of the time in a 400-chunk shared space (200 unrelated cases). Bypassing retrieval: 0.290 -> 0.560.
- Elaborative rehearsal's collapse is architectural, not capacity-limited: swapping in the larger, instruction-tuned flan-t5-base made it worse, not better (0.339 -> 0.212).
- Targeted fixes substantially reduce (not eliminate) the collapse without retraining: on Wiki (n=100), a collapse-detection fallback reached 0.390 and contrastive decoding reached 0.330, vs. 0.130 collapsed.

What this means

- A larger maintenance-rehearsal scorer (RoBERTa-large) improves sentence-selection ranking quality (MAP 0.493 -> 0.584), but downstream QA gains are genre-dependent, not uniform.
- Together, these separate a capacity-bound failure (maintenance rehearsal: bigger helps) from a structural one (elaborative rehearsal: bigger doesn't) -- the central capacity-vs-structure question this talk raises.
- Full-scale validation of the collapse-fix across all four genres remains future work.

Fit over depth.

Not how deeply you process a document — how well the processing fits its structure.

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github.com/lucytheboss/rehearse-then-recall